

AI-based Web Application development and deployment



Estimated Time: 60 minutes

Overview

In this project, we make use of the embedded Watson AI libraries, to create an application that would perform sentiment analysis on a provided text. We then deploy the said application over the web using Flask framework.

Project guidelines

For the completion of this project, you'll have to complete the following 8 tasks, based on the knowledge you have gained through the course.

Tasks and objectives:

- Task 1: Clone the project repository
- Task 2: Create a sentiment analysis application using Watson NLP library
- Task 3: Format the output of the application
- Task 4: Package the application
- Task 5: Run Unit tests on your application
- Task 6: Deploy as web application using Flask

- Task 7: Incorporate Error handling
- Task 8: Run static code analysis

Let's get started !

About Embeddable Watson AI libraries

In this project, you'll be using embeddable libraries to create an AI powered Python application.

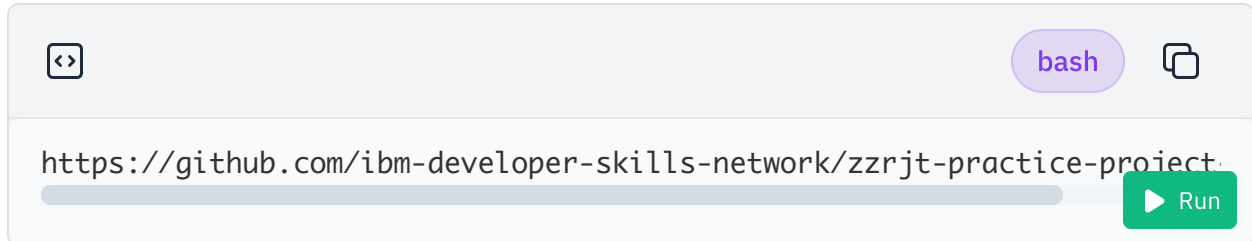
[Embeddable Watson AI libraries](#) include the NLP library, the text-to-speech library and the speech-to-text library. These libraries can be embedded and distributed as part of your application. For your convenience, these libraries have been pre-installed on Skills Network Labs Cloud IDE for use in this project.

The NLP library includes functions for sentiment analysis, emotion detection, text classification, language detection, etc. among others. The speech-to-text library contains functions that perform the transcription service and generates written text from spoken audio. The text-to-speech library generates natural sounding audio from written text.

Please run this lab in the Skills Network Theia Lab environment itself, not in your local IDE (like VS Code). The API used here is hosted on the Skills Network platform and is only accessible within the Theia Lab.

Task 1: Clone the project repository

The Github repository of the project is available on the URL mentioned below.

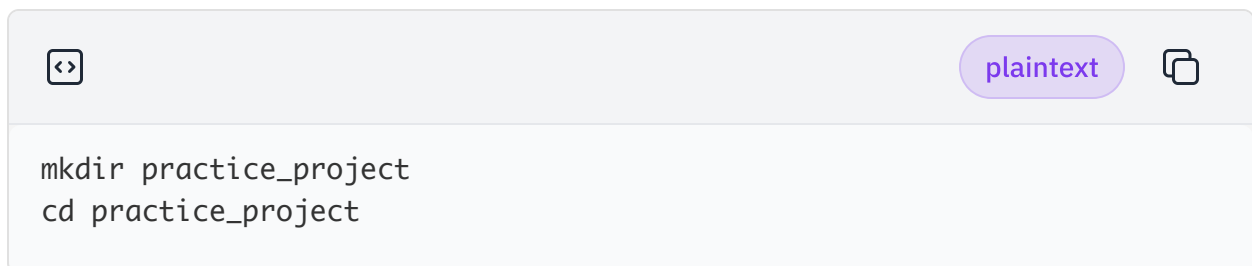


A terminal window with a light gray background. The top bar has a code icon on the left, a 'bash' label in a purple pill, and a copy icon on the right. The main area contains the URL `https://github.com/ibm-developer-skills-network/zzrjt-practice-project`. A green 'Run' button with a play icon is at the bottom right.

You can find instructions on how to fork the repository by visiting exercise 2 in this [Hands-on Lab: Get familiar with Git Commands](#).

Note: Ensure your forked repository is Public.

1. Open a new Terminal and make the directory `practice_project` using `mkdir` command and change the current directory `practice_project` using `cd` command



A terminal window with a light gray background. The top bar has a code icon on the left, a 'plaintext' label in a purple pill, and a copy icon on the right. The main area contains the commands `mkdir practice_project` and `cd practice_project`.

2. Clone this forked GitHub repository using the Cloud IDE terminal to your project to a folder named `practice_project`. You can find instructions on how to get the clone URL of the repository by visiting exercise 3 in this [Hands-on Lab: Get familiar with Git Commands](#).

► Click here for hint

► Click here for solution

- After the cloning is complete, use the terminal to change the current directory `practice_project` using `cd` command.

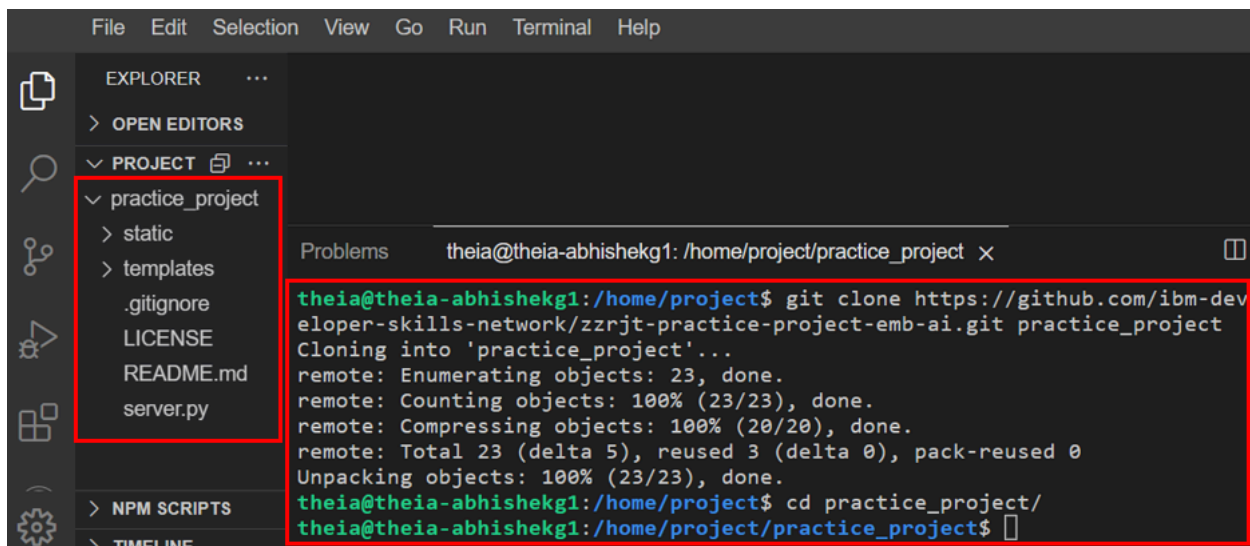
- Ensure Python 3.11 and required libraries are available:

```
python3.11 -V
pip3.11 show requests flask pylint
```

- Install missing libraries:

```
python3.11 -m pip install requests flask pylint
```

- Upon completion, the project tab should have the folder structure as shown in the image.



Task 2: Create a sentiment analysis application using Watson NLP library

NLP sentiment analysis is the practice of using computers to recognize sentiment or emotion expressed in a text. Through NLP, sentiment analysis categorizes words as positive, negative or neutral.

Sentiment analysis is often performed on textual data to help businesses monitor brand and product sentiment in customer feedback, and understanding customer needs. It helps attain the attitude and mood of the wider public which can then help gather insightful information about the context.

For creating the sentiment analysis application, we'll be making use of the Watson Embedded AI Libraries. Since the functions of these libraries are already deployed on the Cloud IDE server, there is no need of importing these libraries to our code. Instead, we need to send a POST request to the relevant model with the required text and the model will send the appropriate response.

A sample code for such an application could be

```
 python   
  
import requests  
  
def <function_name>(<input_args>):  
    url = '<relevant_url>  
    headers = {<header_dictionary>  
    myobj = {<input_dictionary_to_the_function>  
    response = requests.post(url, json = myobj, headers=header)  
    return response.text
```

Note: The response of the Watson NLP functions is in the form of an object. For accessing the details of the response, we can use `text` attribute of the object by calling `response.text` and make the function return the response as simple text.

For this project, you'll be using the BERT based Sentiment Analysis function of the Watson NLP Library. For accessing this function, the URL, the headers and the input json format is as follows.

plaintext

```
URL: 'https://sn-watson-sentiment-bert.labs.skills.network/v1/watson.n
Headers: {"grpc-metadata-mm-model-id": "sentiment_aggregated-bert-work
Input json: { "raw_document": { "text": text_to_analyse } }
```

Here, `text_to_analyse` is being used as a variable that holds the actual written text which is to be analyzed.

In this task, you need to create a new file named `sentiment_analysis.py` in `practice_project` folder. In this file, write the function for running sentiment analysis using the Watson NLP BERT Sentiment Analysis function, as discussed above. Let us call this function `sentiment_analyzer`. Assume that that text to be analysed is passed to the function as an argument and is stored in the variable `text_to_analyse`.

► [Click here for the solution](#)

This application can now be called using Python shell. To test the application, open a Python shell using python3.11 to open the python shell on the current directory i.e `practice_project`. *Make sure that the current directory is `practice_project`.*

bash

```
python3.11
```

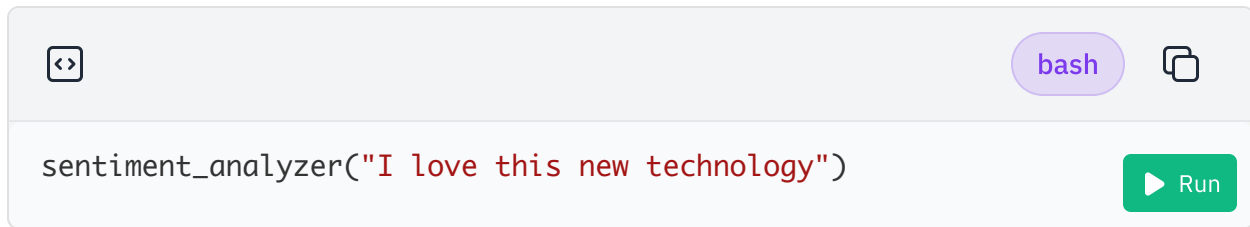
 Run

In the python shell, import the function `sentiment_analyzer`.

► [Click here for hint](#)

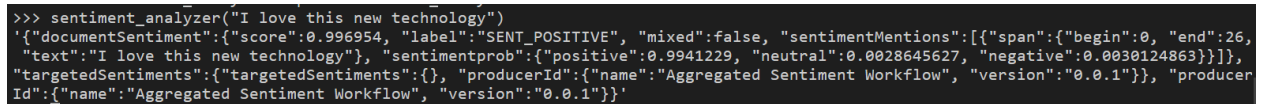
► [Click here for solution](#)

After successful import, test your application with the text "I love this new technology."



The image shows a code editor interface. At the top, there is a toolbar with a code icon (two arrows pointing out of a box), a purple button labeled "bash", and a copy icon (two overlapping squares). Below the toolbar, the code editor contains the text `sentiment_analyzer("I love this new technology")`. To the right of the code, there is a green button with a play icon and the text "Run".

The result expected is as shown in the image below. To exit the python shell, press **Ctrl+Z** or type **exit()**.



```
>>> sentiment_analyzer("I love this new technology")
{'documentSentiment': {'score': 0.996954, 'label': 'SENT_POSITIVE', 'mixed': false, 'sentimentMentions': [{'span': {'begin': 0, 'end': 26, 'text': 'I love this new technology'}, 'sentimentprob': {'positive': 0.9941229, 'neutral': 0.0028645627, 'negative': 0.0030124863}}]}, 'targetedSentiments': {'targetedSentiments': {}, 'producerId': {'name': 'Aggregated Sentiment Workflow', 'version': '0.0.1'}}, 'producerId': {'name': 'Aggregated Sentiment Workflow', 'version': '0.0.1'}}
```

This completes the Task 2. Note that in the output, the information relevant to us is only the **label** and the **score**. In the following task, you will extract this information from this output.

Task 3: Format the output of the application

The output of the application created is in the form of a dictionary, but has been formatted as a text. To access relevant pieces of information from this output, we need to first convert this text into a dictionary. Since dictionaries are the default formatting system for JSON files, we make use of the in-built Python library `json`.

Let's see how this works.

First, in a Python shell, import the json library.

python

```
import json
```

Next, run the `sentiment_analyzer` function for the text "I love this new technology", just like in Task 2, and store the output in a variable called `response`.

python

```
from sentiment_analysis import sentiment_analyzer  
response = sentiment_analyzer("I love this new technology")
```

Now, pass the `response` variable as an argument to the `json.loads` function and save the output in `formatted_response`. Print `formatted_response` to see the difference in the formatting.

python

```
formatted_response = json.loads(response)  
print(formatted_response)
```

```

theia@theia-abhishekg1:/home/project/practice_project$ python3.11
Python 3.11.2 (main, Feb 8 2023, 14:49:25) [GCC 11.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import json
>>> from sentiment_analysis import sentiment_analyzer
>>> response = sentiment_analyzer("I love this new technology")
>>> formatted_response = json.loads(response)
>>> formatted_response
{'documentSentiment': {'score': 0.996954, 'label': 'SENT_POSITIVE', 'mixed': False, 'sentimentMentions': [{span': {'begin': 0, 'end': 26, 'text': 'I love this new technology'}, 'sentimentprob': {'positive': 0.9941229, 'neutral': 0.0028645627, 'negative': 0.0030124863}}]}, 'targetedSentiments': {'targetedSentiments': {}, 'producerId': {'name': 'Aggregated Sentiment Workflow', 'version': '0.0.1'}}, 'producerId': {'name': 'Aggregated Sentiment Workflow', 'version': '0.0.1'}}
>>>

```

Note that the absence of single quotes on either side of the response indicates that this is no longer a text, but is a dictionary instead. To access the correct information from this dictionary, we need to access the keys appropriately. Since this is a nested dictionary structure, i.e. a dictionary of dictionaries, the following statements need to be used to get the label and the score outputs from this response.



python



```

label = formatted_response['documentSentiment']['label']
score = formatted_response['documentSentiment']['score']

```

Check the contents of **label** and **score** to verify the output.

```

>>> label= formatted_response['documentSentiment']['label']
>>> score= formatted_response['documentSentiment']['score']
>>> label
'SENT_POSITIVE'
>>> score
0.996954

```

Now, for Task 3, incorporate the above mentioned technique and make changes to the **sentiment_analysis.py** file. The expected output from calling the **sentiment_analyzer** function should now be a dictionary with 2 keys, label and score, each having the appropriate value extracted from the response of the Watson NLP function. Verify your changes by testing the modified function in a python shell.

► [Click here for the solution](#)

At completion, the expected output of the function is shown in the image below.

```
theia@theia-abhishekg1:/home/project/practice_project$ python3.11
Python 3.11.2 (main, Feb 8 2023, 14:49:25) [GCC 11.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from sentiment_analysis import sentiment_analyzer
>>> sentiment_analyzer("I love this new technology")
{'label': 'SENT_POSITIVE', 'score': 0.996954}
```

To exit the python shell, type `exit()` or press `Ctrl+Z`.

Task 4: Package the application

In this task, you have to package the final application you created in tasks 2 and 3

Let's keep the name of the package as **SentimentAnalysis**. The steps involved in packaging are:

1. Create a folder in the working directory, with the name as the package name.

▶ Click here for hint

▶ Click here for solution

2. Move the application code (i.e., the module) into the package folder.

▶ Click here for hint

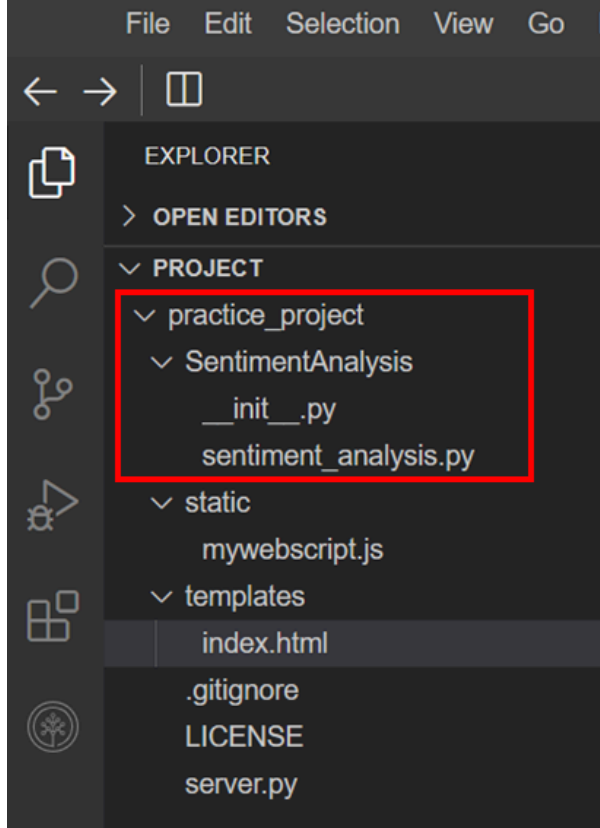
▶ Click here for solution

3. Create the new file as `__init__.py` file, inside the package folder to reference the module.

▶ Click here for hint

▶ Click here for solution

The final folder structure should look as shown in the image below.



SentimentAnalysis is now a valid package and can be imported into any file in this project.

To test this, run a python shell in the terminal and try importing the **sentiment_analyzer** function from the package.

► [Click here for the hint](#)

► [Click here for the solution](#)

No error message received after import statement would indicate that the package is now ready for usage. Test the function by running the following statement in the shell.

 [plaintext](#) 

```
sentiment_analyzer("This is fun.")
```

The output received would look as shown below.

```
>>> sentiment_analyzer("This is fun.")  
{'label': 'SENT_POSITIVE', 'score': 0.997183}
```

To exit the python shell, type `exit()` or press `Ctrl+Z`.

Task 5: Run Unit tests on your application

Since now we have a functional application, it is required that we run unit tests on some test cases to check the validity of its outputs.

For running unit tests, we need to create a new file that calls the required application function from the package and tests its for a known text and output pair.

For this, complete the following steps.

1. Create a new file in `practice_project` folder, called `test_sentiment_analysis.py`.
2. In this file, import the `sentiment_analyzer` function from the `SentimentAnalysis` package. Also import the `unittest` library.

► [Click here for the solution](#)

3. Create the unit test class. Let's call it `TestSentimentAnalyzer`. Define `test_sentiment_analyzer` as the function to run the unit tests.

► [Click here for the solution](#)

4. Define 3 unit tests in the said function and check for the validity of the following statement - label pairs. "I love working with Python": "SENT_POSITIVE" "I hate working with Python": "SENT_NEGATIVE" "I am neutral on Python": "SENT_NEUTRAL"

► [Click here for hint](#)

► [Click here for solution](#)

5. Call the unit tests.

► [Click here for solution](#)

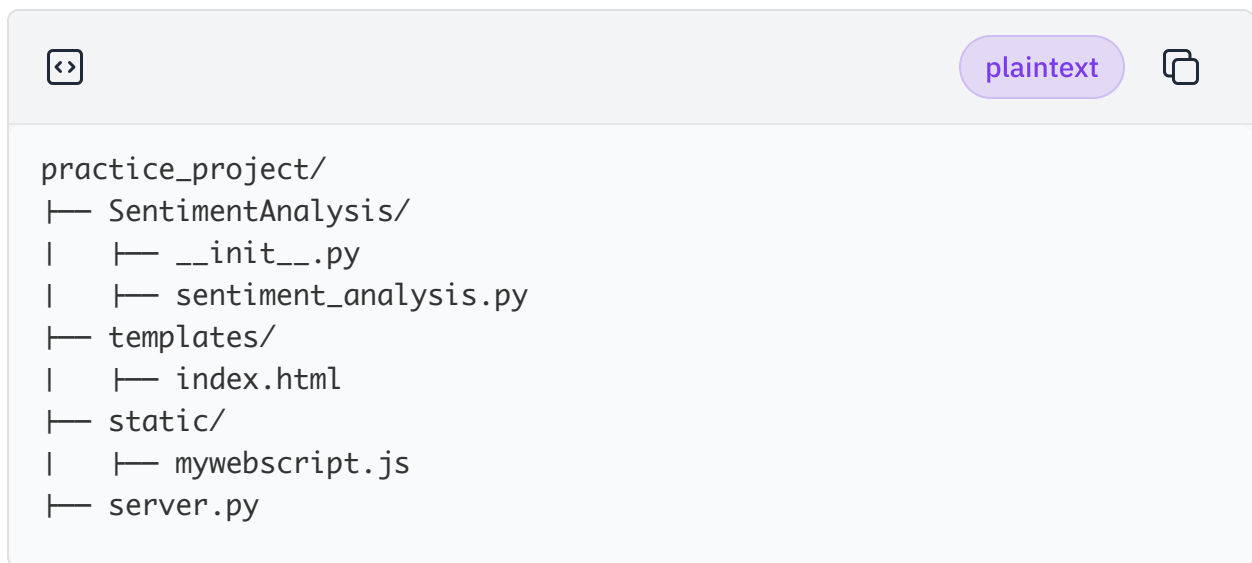
Now that the file is ready, execute the file to perform unit tests. Upon successful execution, the output of this file should be as shown in the image below.

```
theia@theia-abhishekg1:/home/project/practice_project$ python3.11 test_sentiment_analysis.py
.  
-----  
Ran 1 test in 0.501s  
OK
```


Task 6: Deploy as web application using Flask

Now that the application is ready, it is time to deploy it for usage over a web interface. To ease the process of deployment, you have been provided with 3 files which are going to be used for this task.

- Verify directory structure:



The image shows a code editor window with a light gray header. On the left is a code icon, on the right is a 'plaintext' label and a copy icon. The main area displays a directory tree for 'practice_project/':

```
practice_project/  
├── SentimentAnalysis/  
│   ├── __init__.py  
│   └── sentiment_analysis.py  
├── templates/  
│   └── index.html  
├── static/  
│   └── mywebscript.js  
└── server.py
```

- This `index.html` in `templates` folder file has the code for the web interface that has been designed for this lab. This is being provided for you in completion and is to be used as is. You are not required to make any changes to this file.

The interface is as shown in the image.

NLP - Sentiment Analysis using BERT

Please enter the text to be analyzed

Run Sentiment Analysis

Result of Sentiment Analysis

- On clicking the **Run Sentiment Analysis** button, on the html interface, calls this mywebscript.js javascript file in **static** folder which executes a GET request and takes the text provided by the user as input. This text, saved in a variable named **textToAnalyze** is then passed on to the server file to be sent to the application. This file is also being provided to you in completion and is expected to be used as is. You are not required to make any changes to this file.
- Open **server.py** in the **practice_project** folder. This task revolves around the completion of this file. You can complete this file by completing the following 5 steps.

a. *Import the relevant libraries and functions* In this file, you'll need the Flask library along with its **render_template** function (for deploying the HTML file) and **request** function (to initiate the GET request from the web page).

You also would need to import the **sentiment_analyzer** function from the **SentimentAnalysis** package.

Add the relevant lines of code, importing the said functions, in **server.py**

► [Click here for solution](#)

b. *Initiate the Flask app by the name* **Sentiment Analyzer**

Put the knowledge gained in Module 2 of this course and add the statement to server.py, that initiates the application and names it **Sentiment Analyzer**.

► [Click here for solution](#)

c. Define the function `sent_analyzer`

The purpose of this function is two fold. First, the function should send a GET request to the HTML interface to receive the input text. Note that the GET request should reference `textToAnalyze` variable as defined in the `mywebscript.js` file. Store the incoming text to a variable `text_to_analyze`. Now, as the second function, call your `sentiment_analyzer` application with `text_to_analyze` as the argument.

Also, format the returning output of the function in a formal text. For e.g. **The given text has been identified as POSITIVE with a score of 0.99765.**

► [Click here for hint](#)

► [Click here for solution](#)

Note: The function uses the Flask decorator `@app.route("/sentimentAnalyzer")` as referenced in the `mywebscript.js` file.

d. Render the HTML template using `render_index_page`

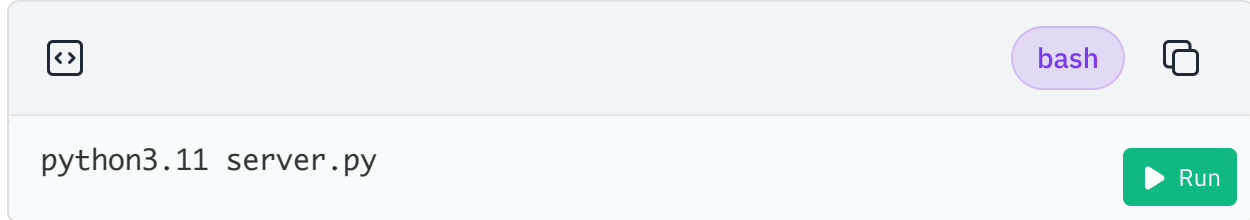
This function should simply run the `render_template` function on the HTML template, `index.html`.

► [Click here for solution](#)

e. Run the application on `localhost:5000`

Finally, upon file execution, run the application on host: `0.0.0.0` (or localhost) on port number 5000.

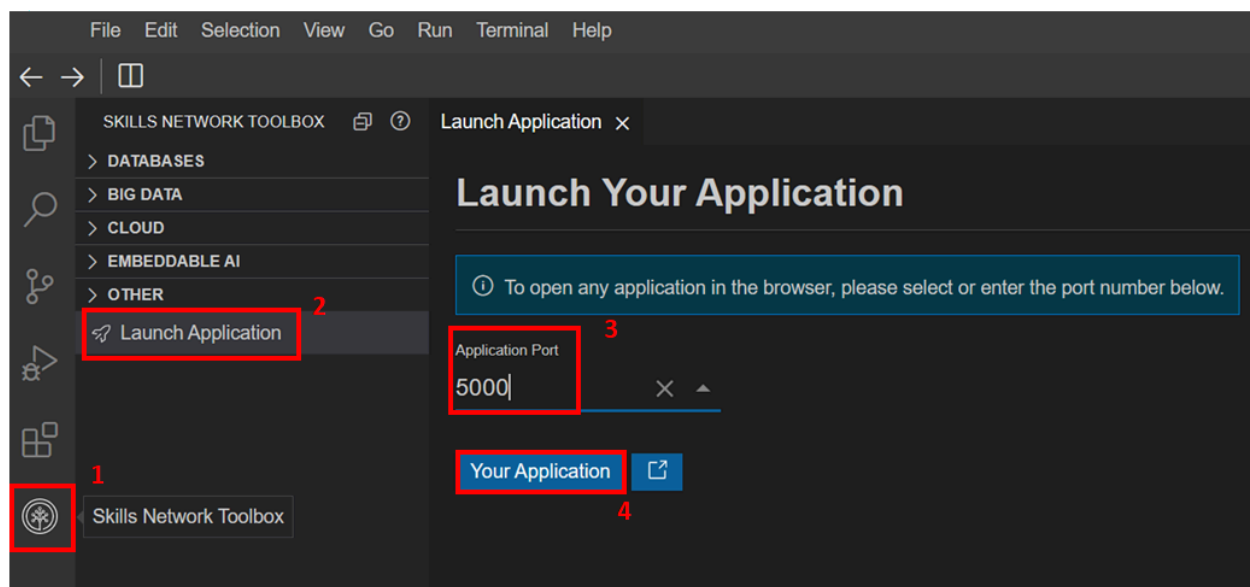
► [Click here for solution](#)



The output would look like this.

```
theia@theia-abhishekg1:/home/project/practice_project$ python3.11 server.py
* Serving Flask app 'Sentiment Analyzer'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.22.138.184:5000
Press CTRL+C to quit
```

The app is now running on localhost:5000. To access the application, go to the Skills Network Toolbox tab and click on Launch Application. Enter the Application port as 5000, and click on Your Application.



The application interface will open. Use the interface to test your application.

server.py sentiment_analysis.py __init__.py Your Application x index.html

< > ↺ https://abhishekg1-5000.theianext-0-labs-prod-misc-tools-us-east-0.proxy.cognitiveclass.ai/

NLP - Sentiment Analysis using BERT

Please enter the text to be analyzed

I love this new technology

Run Sentiment Analysis

Result of Sentiment Analysis

The given text has been identified as POSITIVE with a confidence score of 0.996954.

Problems theia@theia-abhishekg1: /home/project/practice_project x

127.0.0.1 - - [09/Jul/2023 15:03:50] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [09/Jul/2023 15:03:51] "GET /static/mywebscript.js HTTP/1.1" 304 -
127.0.0.1 - - [09/Jul/2023 15:03:55] "GET /sentimentAnalyzer?textToAnalyze=I%20love%20this%20new%20technology HTTP/1.1" 200 -
□

To stop the application, press **Ctrl+C**.

Task 7: Incorporate Error handling

To incorporate error handling, we need to identify the different forms of error codes that may be received in response to the GET query initiated by the `sent_analyzer` function in `server.py`.

This is already a part of the Watson NLP Library functions and can be observed on the terminal console where the code is running.

Consider the image shown below.

NLP - Sentiment Analysis using BERT

Please enter the text to be analyzed

I love this new technology

Run Sentiment Analysis

Result of Sentiment Analysis

The given text has been identified as POSITIVE with a confidence score of 0.996954.

```
Problems theia@theia-abhishek1: /home/project/practice_project x
127.0.0.1 - - [09/Jul/2023 15:03:50] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [09/Jul/2023 15:03:51] "GET /static/mywebscript.js HTTP/1.1" 304 -
127.0.0.1 - - [09/Jul/2023 15:03:55] "GET /sentimentAnalyzer?textToAnalyze=I%20love%20this%20new%20technology HTTP/1.1" 200 -
```

The codes indicate that the initial GET request was successful (200), the request was then returned as Not Modified from the cache (304) and then the GET request to generate the response was also conducted successfully.

In the case of invalid entries, the system responds with 500 error code, indicating that there is something wrong at the server end. Invalid entry could be anything that the model is not able to interpret. However, in the situation of this error, this application output doesn't get updated.

NLP - Sentiment Analysis using BERT

Please enter the text to be analyzed

asda!ka skd;alk a;lksda

Run Sentiment Analysis

Result of Sentiment Analysis

The given text has been identified as POSITIVE with a confidence score of 0.996954.

```
Problems theia@theia-abhishekg1:/home/project/practice_project x
File "/home/project/practice_project/SentimentAnalysis/sentiment_analysis.py", line 10, in sentiment_analyzer
    label = formatted_response['documentSentiment']['label']
KeyError: 'documentSentiment'
127.0.0.1 - - [10/Jul/2023 15:46:39] "GET /sentimentAnalyzer?textToAnalyze=asda!ka%20skd;alk%20a;lksda HTTP/1.1" 500 -
```

Note that the output on the interface is the same as before, the text being analyzed is a random text and the Watson AI libraries are throwing a 500 error confirming that the model has not been able to process the request.

To fix this bug in our application, we need to study the response received from the Watson AI library function, when the server generates 500 error. To test this, we need to retrace the steps taken in Task 2, and test the Watson AI library with an invalid string input.

Open a python shell in the terminal and run the following commands to check the required output after updating sentiment_analysis.py file with below.



python



```
import requests

# Define the URL for the sentiment analysis API
url = "https://sn-watson-sentiment-bert.labs.skills.network/v1/watson.runtime.nlp.v1/NlpService/SentimentPredict"

# Set the headers with the required model ID for the API
headers = {"grpc-metadata-mm-model-id": "sentiment_aggregated-bert-workflow_lang_multi_stock"}

# Define the first payload with nonsensical text to test the API
myobj = { "raw_document": { "text": "as987da-6s2d aweadsa" } }

# Make a POST request to the API with the first payload and headers
response = requests.post(url, json=myobj, headers=headers)

# Print the status code of the first response
print(response.status_code)

# Define the second payload with a meaningful text to test the API
myobj = { "raw_document": { "text": "Testing this application for error handling" } }

# Make a POST request to the API with the second payload and headers
response = requests.post(url, json=myobj, headers=headers)

# Print the status code of the second response
print(response.status_code)
```

The console response looks as shown in the image below. The red boxes indicate the invalid text and its status code received, and the yellow boxes indicate the valid text and its status code received.

```
theia@theia-abhishekg1:/home/project/practice_project$ python3.11
Python 3.11.2 (main, Feb 8 2023, 14:49:25) [GCC 11.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import requests
>>> url = "https://sn-watson-sentiment-bert.labs.skills.network/v1/watson.runtime.nlp.v1/NlpService/SentimentPredict"
>>> headers = {"grpc-metadata-mm-model-id": "sentiment_aggregated-bert-workflow_lang_multi_stock"}
>>> myobj = { "raw_document": { "text": "as987da-6s2d aweadsa" } }
>>> response = requests.post(url, json = myobj, headers=headers)
>>> print(response.status_code)
500
>>>
>>> myobj = { "raw_document": { "text": "Testing this application for error handling" } }
>>> response = requests.post(url, json = myobj, headers=headers)
>>> print(response.status_code)
200
>>>
```


This enables you to modify the application in such a fashion, that we can send different outputs for different status codes.

In the first part of this task, you have to modify the `sentiment_analyzer()` function to return the both `label` and `score` as `None` in case of invalid text entry.

▶ Click here for hint

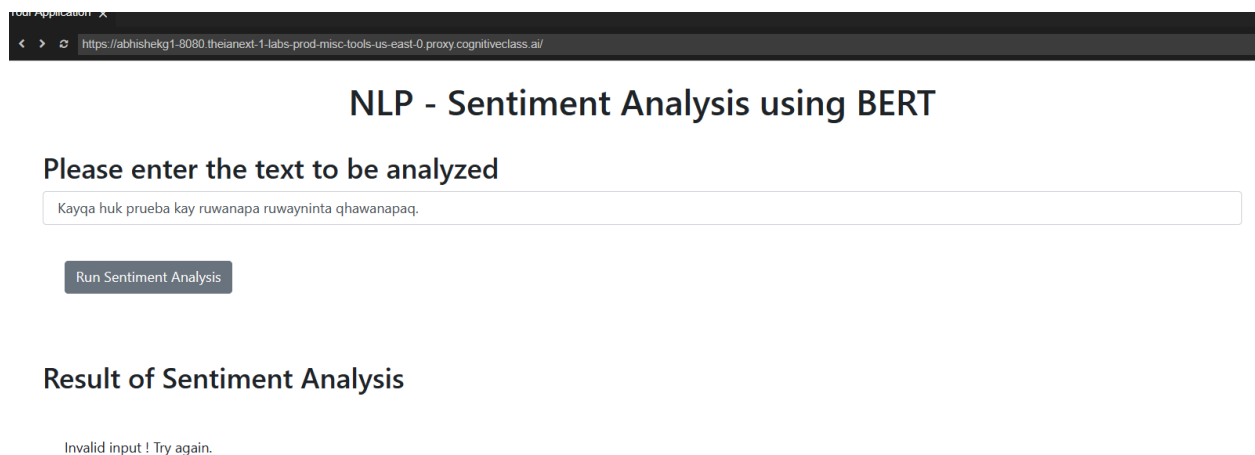
▶ Click here for the solution

Now, in `server.py`, the response to be sent to the console should also be different for the valid and invalid input types. For invalid input, let the console print `Invalid input ! Try again.`

▶ Click here for hint

▶ Click here for the solution

Now, your application is capable of responding appropriately to any form of inputs.



Task 8: Run static code analysis

Finally, in Task 8, we check the quality of your coding skills as per the PEP8 guidelines by running static code analysis.

Normally, this is done at the time of packaging and unit testing the application. However, we have kept this step at the end of this project since the codes are updated in all tasks before this. Once your files for this project are now ready, let us test them for adherence to the PEP8 guidelines. Since **PyLint** was already installed in Task 1, you can proceed directly to running the analysis. If for any reason it is not available in your environment, you can install it using the terminal.

► [Click here for the solution](#)

Next, use pylint to run static code analysis on **server.py**.

► [Click here for the solution](#)

If all aspects of PEP8 guide have been incorporated in your code, then the score generated should be 10/10. In case it isn't, follow the instructions given by pylint library to modify to correct the code appropriately.

```
theia@theia-abhishekg1:/home/project/practice_project$ pylint server.py
-----
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

This concludes the practice project.

(Optional) Additional Exercises

Interested learners can try the following exercises on your own, for enhanced understanding of the concepts learnt through this project. No solution is being provided for these exercises. However, feel free to discuss and share your solutions with your peers in the course discussion forums.

1. Run static code analysis on `sentiment_analysis.py`. Try to achieve a 10/10 score. Hint: *Docstrings*
2. Test the capacity of your application in handling sentences of languages other than English, for e.g. French, German, etc. See if the application responds with an invalid text error.
3. Currently, if the application is run WITHOUT supplying an input, i.e. leaving the text blank, the model still throws the same error of invalid text. Try including a special case, where a blank input receives a different error message.

Conclusion

Congratulations on completing this project.

With the completion of this project, you have:

1. Created an AI based sentiment analysis application using Watson NLP embedded libraries.
2. Formatted the output received from the Watson NLP library function to extract relevant information from it.
3. Packaged the application and made it importable to any python code for usage.
4. Ran unit tests on the application and checked the validity of its outputs for different inputs.
5. Deployed the application using Flask framework.
6. Incorporated error handling capability in the application, such that a response code of 500 receives an appropriate response from the application.
7. Ran static code analysis on the code files to confirm their adherence to the PEP8 guidelines.

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About Embeddable Watson AI libraries

application.

[Embeddable Watson AI libraries](#) include the NLP library, the text-to-speech library and the speech-to-text library. These libraries can be embedded and distributed as part of your application. For your convenience, these libraries have been pre-installed on Skills Network Labs Cloud IDE for use in this project.

The NLP library includes functions for sentiment analysis, emotion detection, text classification, language detection, etc. among others. The speech-to-text library contains functions that perform the transcription service and generates written text from spoken audio. The text-to-speech library generates natural sounding audio from written text.

Please run this lab in the Skills Network Theia Lab environment itself, not in your local IDE (like VS Code). The API used here is hosted on the Skills Network platform and is only accessible within the Theia Lab.

← AI-based Web Application...

Task 1: Clone the project ... →